



Independent University, Bangladesh (IUB)

CSE211L Project:

An Algorithmic System for Structuring Dhaka's Cake Sellers

Group: 06

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Abstract

The "Dhaka Cake Seller Management System" is a simulation-based project that models the distribution, management, and optimization of cake sellers within a virtual Dhaka city environment. The project generates synthetic vendor and road-network data, applies multiple sorting and searching algorithms for vendor ranking, allocates vendors to nodes with capacity constraints, and finally uses the Dijkstra shortest path algorithm to compute optimal travel routes. This 3-milestone system demonstrates practical applications of algorithmic thinking, data visualization, and optimization techniques. The report details the reasoning behind the selection of different sorting algorithms, analyses their performance, and discusses real-world relevance.

Introduction

Urban food vendor management is an important task for improving hygiene, public safety, and consumer convenience. However, assigning hundreds of vendors across a city with limited area capacity and determining optimal travel routes can be computationally challenging.

This project creates an artificial model of Dhaka's streets using a weighted graph, and randomly generates cake seller data including hygiene score, revenue, experience, and location. The goal is to:

1. Sort cake sellers efficiently based on hygiene score.
2. Allocate them to the nearest city nodes with capacity limits.
3. Compute shortest routes between users and selected sellers.
4. Visualize the city map, seller locations, allocations, and routes.

This multi-step workflow shows how algorithmic solutions can solve real-world management problems in smart city systems.

Problem Statement and Proposed Solution

Problem Statement

Dhaka contains many small-scale cake sellers scattered randomly. The challenges include:

- Ensuring high-hygiene vendors are easy to identify.
- Allocating vendors into logical city zones with limited capacity (to avoid overcrowding).
- Helping users find an optimal travel route to the seller they want to visit.
- Visualizing everything on a map for interpretability.

Proposed Solution

We designed a three-milestone algorithmic system:

Milestone 1: Sorting & Searching

We sort vendors based on hygiene score, because this is the most important metric in food vendor selection. Five sorting algorithms were implemented:

- Bubble Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Counting Sort

We also implemented Linear Search to locate a seller by ID.

Performance Discussion

Algorithm	Time	Comparisons	Swaps	Notes
Bubble Sort	Slow	Very High	Very High	Good for teaching, bad for performance
Insertion Sort	Better	Medium	Medium	Efficient for nearly-sorted data
Merge Sort	Fast & Stable	$O(n \log n)$	Medium	Very reliable for large datasets
Quick Sort	Fastest in most cases	$O(n \log n)$ avg	Low	Best practical choice
Counting Sort	Very Fast	Zero comparisons	High memory	Perfect for hygiene scores (0–100 range)

Why We Chose These Sorting Algorithms

We intentionally selected a mixture of simple, intermediate, and efficient algorithms to compare behavior:

1. Bubble Sort

- Easiest to understand; great for demonstrating comparisons/swaps.
- Used for educational benchmarking.

2. Insertion Sort

- Performs well on small or almost-sorted lists.
- Good demonstration of adaptive behavior.

3. Merge Sort

- Always $O(n \log n)$.
- Stable sorting makes it good for real systems.

4. Quick Sort

- Fastest for large datasets in practice.
- Low overhead; excellent for real-world applications.
- We used this sorted list for the remaining milestones.

5. Counting Sort

- Hygiene score is limited to 0–100 → perfect for a non-comparison sort.
- Demonstrates how domain-specific algorithms outperform general ones.

In short, we chose these algorithms to observe and compare performance across different algorithm categories: (basic, adaptive, divide-and-conquer, practical, and counting-based).

Milestone 2: Vendor Allocation to City Nodes

Each city node has a capacity of 5 sellers.

We compute the Euclidean distance from each vendor to every node, sort the distances internally, and assign vendors to the nearest available node.

If no node has remaining space → vendor goes to waitlist.

Results

- Total vendors: 100
- Total node capacity: 75
- Allocated vendors: 75
- Waitlisted vendors: 25

A full table was generated showing node occupancy.

Visualization

Two maps were generated:

1. Before allocation – vendors scattered randomly
2. After allocation – vendors assigned to nodes, with lines showing their relocation

This helps visually validate whether the algorithm correctly groups vendors.

Milestone 3: Shortest Path Finder (Dijkstra's Algorithm)

Users can:

- Input their current node

- Select a seller
- View the shortest path to reach that seller's assigned node

We implemented Dijkstra's Algorithm because:

- It is optimal for non-negative weighted graphs
- It guarantees shortest path
- It runs efficiently with priority queues
- It fits real-world road networks perfectly

Limitations

Although the system performs well in simulation, limitations include:

- Vendor locations are random, not real-world accurate.
- Road network is artificially generated, not based on Dhaka's actual GIS data.
- Capacity limit per node is fixed (5), which may not reflect real zoning policies.
- Linear Search is slow for large datasets; Binary Search would be better after sorting.
- Dijkstra works well, but large real-world networks would require more advanced algorithms like A*.
- No real-time data update or dynamic vendor movement.

Conclusion and Future Work

The Dhaka Cake Seller Management System successfully demonstrates how algorithmic approaches can solve major challenges in urban vendor management:

- Sorting vendors by hygiene ensures transparency and public safety
- Capacity-aware allocation prevents overcrowding
- Shortest-path computation improves navigation and accessibility
- Full visualizations support decision-making and debugging

Future Work

To extend this project:

1. Integrate real GPS & GIS map data from Dhaka
2. Add A* pathfinding for faster route prediction
3. Build a web dashboard with live map rendering
4. Implement clustering (k-means) to create zones dynamically
5. Add vendor rating, customer reviews, and time-based popularity
6. Introduce machine-learning predictions
7. Add load balancing between nodes depending on demand

References

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